

AI-Enabled IoT Smart Energy Meter

with Historical Energy Analysis and Next-Day Consumption Prediction

Review stage Zeroth / Preliminary Review	Project status Planning and technical study phase
Core platform ESP32-based energy monitoring	Planned intelligence Data analytics and load forecasting

Purpose of this review

This document records the initial problem study, proposed system design, candidate data sources, and implementation roadmap. Hardware assembly, live data collection, dashboard development, and trained forecasting results are planned next-stage activities and are not represented as completed work.

Prepared for departmental preliminary review

1. Title and Introduction

Electricity use in homes and small buildings is usually visible only as a monthly bill. That makes it difficult to identify peak-use periods, understand consumption behaviour, or act before avoidable energy cost occurs. The proposed project combines low-cost IoT metering with a web-based analytics interface and short-term energy forecasting.

Proposed title

AI-Enabled IoT Smart Energy Meter with Predictive Energy Analytics Using ESP32.

Proposed solution

An ESP32-connected sensing unit will acquire electrical measurements and transmit timestamped readings through Wi-Fi. A database and dashboard will store, visualise, and analyse the data. A forecasting module will use historical consumption patterns to estimate the next day's energy consumption in kWh.

Why this project

The project joins embedded systems, IoT communication, web development, database design, data analysis, and machine learning in one demonstrable application. The aim is not only to display readings, but to turn them into useful information and a practical next-day forecast.

Current preliminary work

The initial scope, measurement fields, software architecture, and public benchmark datasets have been identified. The proposed hardware reference is the ESP32 energy-metering setup selected by the project team. Component validation and integration are scheduled after the preliminary review.

2. Objectives and Proposed Workflow

Objective	Expected output
Acquire electrical readings	Timestamped voltage, current, power and energy values
Analyse historical consumption	Hourly/daily charts, peak-use periods, estimated cost
Forecast next-day demand	Predicted daily kWh and comparison with actual value
Notify abnormal use	Threshold-based high-consumption alert

End-to-end proposed workflow

1. Energy sensing	2. ESP32 + Wi-Fi	3. Database	4. Analytics dashboard	5. Next-day forecast
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Measurement and database plan

Each device record will include device ID, timestamp, voltage (V), current (A), active power (W), cumulative energy (kWh), and optional estimated cost. Readings will be transmitted every 1-5 minutes and aggregated into hourly and daily totals for analysis and forecasting.

Safety note: work involving mains electricity must be assembled and tested only under qualified supervision, with suitable isolation and enclosures.

3. Existing Systems and Proposed Contribution

Many low-cost smart meters provide either a local display or simple remote monitoring. Their limitation is that they commonly stop at showing the current reading. The proposed project adds a data layer that makes consumption understandable over time and supports a next-day forecast.

Typical existing meter	Proposed project
Displays instantaneous units or power	Stores timestamped historical readings
Limited usage interpretation	Shows trends, peaks, daily totals and cost estimate
No prediction	Predicts next-day consumption using historical features
Local/device-centred	IoT dashboard accessible through web/mobile interface

Forecasting approach under study

The first baseline will use the previous day's consumption as the next-day estimate. Candidate ML models are Linear Regression and Random Forest Regression. Features will include lagged daily consumption, day of week, weekend flag, recent moving average, and optional temperature/weather data. Accuracy will be reported with MAE and RMSE after project data is available.

Planned stack: ESP32 firmware; Wi-Fi/MQTT or REST transmission; Firebase or a Python API with database; React/Flutter dashboard; Python, Pandas and scikit-learn for analysis.

4. Dataset Exploration and Initial Data Plan

Before locally generated readings are available, public datasets have been shortlisted to understand data structure, select forecasting features, and prototype the analysis pipeline. These datasets are references for preliminary model development; they are not presented as data collected by this project.

Dataset / source	Relevant characteristics	Planned use
UCI Appliances Energy Prediction	19,735 observations; 10-minute appliance energy readings; temperature, humidity and weather variables; 4.5 months.	Prototype preprocessing and regression experiments.
UCI ElectricityLoadDiagrams20112014	370 clients; 15-minute electricity values; multi-year time series.	Study aggregation and next-day load forecasting.
Project-generated ESP32 data (planned)	Voltage, current, power, energy and timestamp at 1-5 minute intervals.	Final dashboard, model training and evaluation.

Candidate analysis outputs

Daily total kWh; hourly average load profile; highest-consumption day; rolling seven-day average; comparison of predicted versus actual daily use; and alert when consumption exceeds an adjustable threshold.

Reference links

1. UCI Appliances Energy Prediction: <https://archive-beta.ics.uci.edu/dataset/374/appliances%2Benergy%2Bprediction>
2. UCI ElectricityLoadDiagrams20112014: <https://archive.ics.uci.edu/dataset/321/electricityloadaddiagrams20112014>
3. UCI Machine Learning Repository: <https://archive.ics.uci.edu/>

5. Work Plan, Expected Outcome and Next Steps

Phase	Planned activity	Evidence / deliverable
1. Hardware validation	Verify sensing readings and ESP32 Wi-Fi transmission.	Circuit record and sample timestamped data.
2. Data platform	Create data schema, ingestion API and dashboard screens.	Live readings and history charts.
3. Analytics	Aggregate data and identify trends, peaks and anomalies.	Analysis report and visualisations.
4. ML forecast	Train baseline and regression model with time-based features.	Next-day forecast with MAE/RMSE comparison.
5. Integration	Connect final device, dashboard and forecast module.	End-to-end working demonstration.

Expected outcome

The completed system is expected to provide a low-cost IoT energy-monitoring solution that records consumption data, explains historical usage through an accessible dashboard, and estimates the next day's electricity use. The final evaluation will compare forecasted and actual daily consumption to quantify accuracy.

Thank you